

Sixth Section.

Multiplication and Division of the Elliptic Functions.

§ 57. Multiplication of the elliptic functions.

By the multiplication of the elliptic functions one understands the representation of the functions $\operatorname{sn} nv$, $\operatorname{cn} nv$, $\operatorname{dn} nv$, for an integral n , as rational functions of $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$; this is a problem which, as is clear from the addition theorem, can always be solved.

The form of the solution follows easily from the consideration of the ϑ -functions.

As is seen immediately from § 21, the functions $\vartheta_{g_1, g_2}(nu)$ are ϑ -functions of order n^2 , whose characteristic is equal to $(0, 0)$ when n is even, and equal to (g_1, g_2) when n is odd. Moreover $\vartheta_{11}(nu)$ is an odd function of u , whereas $\vartheta_{00}(nu)$, $\vartheta_{10}(nu)$, $\vartheta_{01}(nu)$ are even functions of u . These functions can therefore be represented rationally by the ϑ -functions, and the theorems of § 21 give the form of these expressions. If we again denote by $F^{(\nu)}(x, y)$ an integral rational homogeneous function of order ν , we obtain:

for even n :

$$\begin{aligned}\vartheta_{11}(nu) &= \vartheta_{00}(u)\vartheta_{01}(u)\vartheta_{10}(u)\vartheta_{11}(u)F^{\frac{n^2-4}{2}}\left[\vartheta_{11}^2(u), \vartheta_{01}^2(u)\right], \\ \vartheta_{g_1, g_2}(nu) &= F^{\frac{n^2}{2}}\left[\vartheta_{11}^2(u), \vartheta_{01}^2(u)\right], \quad (g_1, g_2) = (10), (01), (00),\end{aligned}\tag{1}$$

for odd n :

$$\vartheta_{g_1, g_2}(nu) = \vartheta_{g_1, g_2}(u)F^{\frac{n^2-1}{2}}\left[\vartheta_{11}^2(u), \vartheta_{01}^2(u)\right].\tag{2}$$

If we divide these formulae by $\vartheta_{01}(u)^{n^2}$, then the right hand sides can be represented as integral rational functions of the elliptic functions $\operatorname{sn} v$, $\operatorname{cn} v$, $\operatorname{dn} v$ (§ 42). Thus, putting

$$v = \pi\vartheta_{00}^2 u, \quad \operatorname{sn} v = x, \quad \operatorname{cn} v = y, \quad \operatorname{dn} v = z,\tag{3}$$

we can write (1), (2), after suitable determination of a constant factor, as follows:

I. for even n :

$$\begin{aligned}\frac{\vartheta_{01}^{n^2}\vartheta_{00}}{\vartheta_{10}} \frac{\vartheta_{11}(nu)}{\vartheta_{01}(u)^{n^2}} &= xyz A(x^2), \quad A(0) = n, \\ \frac{\vartheta_{01}^{n^2}}{\vartheta_{10}} \frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}} &= B(x^2), \quad B(0) = 1, \\ \frac{\vartheta_{01}^{n^2}}{\vartheta_{00}} \frac{\vartheta_{00}(nu)}{\vartheta_{01}(u)^{n^2}} &= C(x^2), \quad C(0) = 1, \\ \vartheta_{01}^{n^2-1} \frac{\vartheta_{01}(nu)}{\vartheta_{01}(u)^{n^2}} &= D(x^2), \quad D(0) = 1.\end{aligned}$$

II. for odd n :

$$\begin{aligned}\frac{\vartheta_{00}\vartheta_{01}^{n^2-1}}{\vartheta_{10}} \frac{\vartheta_{11}(nu)}{\vartheta_{01}(u)^{n^2}} &= x A(x^2), \quad A(0) = n, \\ \frac{\vartheta_{01}^{n^2}}{\vartheta_{10}} \frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}} &= y B(x^2), \quad B(0) = 1, \\ \frac{\vartheta_{01}^{n^2}}{\vartheta_{00}} \frac{\vartheta_{00}(nu)}{\vartheta_{01}(u)^{n^2}} &= z C(x^2), \quad C(0) = 1, \\ \vartheta_{01}^{n^2-1} \frac{\vartheta_{01}(nu)}{\vartheta_{01}(u)^{n^2}} &= D(x^2), \quad D(0) = 1,\end{aligned}$$

where A, B, C, D are integral rational functions of x^2 , whose degrees are obtained from (1), (2) as follows:

	$A(x^2)$	$B(x^2)$	$C(x^2)$	$D(x^2)$	
degree:	$\frac{n^2}{2} - 2$	$\frac{n^2}{2}$	$\frac{n^2}{2}$	$\frac{n^2}{2}$	n even,
	$\frac{n^2-1}{2}$	$\frac{n^2-1}{2}$	$\frac{n^2-1}{2}$	$\frac{n^2-1}{2}$	n odd.

From the definitions I, II one immediately obtains the values of $A(0), B(0), C(0), D(0)$, that is, the terms independent of n , as they have been added above.

Numerous relations between these functions are still obtained in the following way. To carry out one example here somewhat more fully, replace v by $v + K$, and therefore u by $u + \frac{1}{2}$. Then x, y, z go, by § 44, (18), into $y/z, -\kappa'x/z, \kappa'/z$, and by § 21, (8), from the first formula II for an odd n one obtains

$$\frac{\vartheta_{00}\vartheta_{01}^{n^2-1}}{\vartheta_{10}} \frac{\vartheta_{10}(nu)}{\vartheta_{01}(u)^{n^2}} \left(\frac{\vartheta_{01}(u)}{\vartheta_{00}(u)} \right)^{n^2} = (-1)^{\frac{n-1}{2}} \frac{y}{z} A\left(\frac{y^2}{z^2}\right).$$

On the other hand, by the second equation II and by § 42, the left hand side is equal to

$$\frac{y}{z} \left(\frac{\sqrt{\kappa'}}{z} \right)^{n^2-1} B(x^2),$$

and consequently

$$B(x^2) = (-1)^{\frac{n-1}{2}} \left(\frac{z}{\sqrt{\kappa'}} \right)^{n^2-1} A\left(\frac{y^2}{z^2}\right).$$

Putting in this $x = 1, y = 0, z = \kappa'$, or $x = 0, y = 1, z = 1$, gives

$$B(1) = (-1)^{\frac{n-1}{2}} n \sqrt{\kappa'}^{n^2-1}, \quad A(1) = (-1)^{\frac{n-1}{2}} \sqrt{\kappa'}^{n^2-1}.$$

One can derive a whole system of such formulae by making in I and II the following substitutions:

u	x	y	z
$u + \frac{1}{2}$	$\frac{y}{z}$	$-\frac{\kappa'x}{z}$	$\frac{\kappa'}{z}$
$u + \frac{\omega}{2}$	$\frac{1}{\kappa x}$	$-\frac{iz}{\kappa x}$	$-\frac{iy}{x}$
$u + \frac{1+\omega}{2}$	$\frac{z}{\kappa y}$	$\frac{i\kappa'}{\kappa y}$	$\frac{i\kappa'x}{y}$.

Thus one obtains:

$$\left(\frac{z}{\sqrt{\kappa'}} \right)^{n^2-4} A\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n}{2}-1} A(x^2), \quad A(1) = (-1)^{\frac{n}{2}-1} n \sqrt{\kappa'}^{n^2-4}, \quad (4a)$$

$$\left(\frac{z}{\sqrt{\kappa'}} \right)^{n^2} B\left(\frac{y^2}{z^2}\right) = (-1)^{\frac{n}{2}} B(x^2), \quad B(1) = (-1)^{\frac{n}{2}} \sqrt{\kappa'}^{n^2},$$

$$\left(\frac{z}{\sqrt{\kappa'}} \right)^{n^2} C\left(\frac{y^2}{z^2}\right) = C(x^2), \quad C(1) = \sqrt{\kappa'}^{n^2},$$

$$\left(\frac{z}{\sqrt{\kappa'}} \right)^{n^2} D\left(\frac{y^2}{z^2}\right) = D(x^2), \quad D(1) = \sqrt{\kappa'}^{n^2}, \quad n \text{ even.}$$

Likewise,

$$\begin{aligned}
\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} A\left(\frac{y^2}{z^2}\right) &= (-1)^{\frac{n-1}{2}} B(x^2), & A(1) &= (-1)^{\frac{n-1}{2}} \sqrt{\varkappa'}^{n^2-1}, \\
\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} B\left(\frac{y^2}{z^2}\right) &= (-1)^{\frac{n-1}{2}} A(x^2), & B(1) &= (-1)^{\frac{n-1}{2}} n \sqrt{\varkappa'}^{n^2-1}, \\
\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} C\left(\frac{y^2}{z^2}\right) &= D(x^2), & C(1) &= \sqrt{\varkappa'}^{n^2-1}, \\
\left(\frac{z}{\sqrt{\varkappa'}}\right)^{n^2-1} D\left(\frac{y^2}{z^2}\right) &= C(x^2), & D(1) &= \sqrt{\varkappa'}^{n^2-1},
\end{aligned} \tag{5}$$

for n odd.

Further,

$$\begin{aligned}
(\sqrt{\varkappa} x)^{n^2-4} A\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n}{2}-1} A(x^2), & A(\infty) &= (-1)^{\frac{n}{2}-1} n (\sqrt{\varkappa})^{n^2-4}, \\
(\sqrt{\varkappa} x)^{n^2} B\left(\frac{1}{\varkappa^2 x^2}\right) &= B(x^2), & B(\infty) &= \sqrt{\varkappa}^{n^2}, \\
(\sqrt{\varkappa} x)^{n^2} C\left(\frac{1}{\varkappa^2 x^2}\right) &= C(x^2), & C(\infty) &= \sqrt{\varkappa}^{n^2}, \\
(\sqrt{\varkappa} x)^{n^2} D\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n}{2}} D(x^2), & D(\infty) &= (-1)^{\frac{n}{2}} \sqrt{\varkappa}^{n^2},
\end{aligned} \tag{6}$$

for n even, and

$$\begin{aligned}
(\sqrt{\varkappa} x)^{n^2-1} A\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n-1}{2}} D(x^2), & A(\infty) &= (-1)^{\frac{n-1}{2}} \sqrt{\varkappa}^{n^2-1}, \\
(\sqrt{\varkappa} x)^{n^2-1} B\left(\frac{1}{\varkappa^2 x^2}\right) &= C(x^2), & B(\infty) &= \sqrt{\varkappa}^{n^2-1}, \\
(\sqrt{\varkappa} x)^{n^2-1} C\left(\frac{1}{\varkappa^2 x^2}\right) &= B(x^2), & C(\infty) &= \sqrt{\varkappa}^{n^2-1}, \\
(\sqrt{\varkappa} x)^{n^2-1} D\left(\frac{1}{\varkappa^2 x^2}\right) &= (-1)^{\frac{n-1}{2}} A(x^2), & D(\infty) &= (-1)^{\frac{n-1}{2}} n \sqrt{\varkappa}^{n^2-1},
\end{aligned} \tag{7}$$

for n odd.

Here $A(\infty), B(\infty), C(\infty), D(\infty)$ are in each case to mean the coefficients of the highest power of x in these functions. In (7) the two last formulae can be derived from the first by interchanging x with $1/\varkappa x$.

The third system of formulae can be obtained either in the same way, or also by replacing x by $1/\varkappa x$ in (4), (5) and applying (6), (7):

$$\begin{aligned}
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2-4} A\left(\frac{z^2}{\varkappa^2 y^2}\right) &= A(x^2), & A\left(\frac{1}{\varkappa^2}\right) &= n \sqrt{\frac{\varkappa'}{\varkappa}}^{n^2-4}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2} B\left(\frac{z^2}{\varkappa^2 y^2}\right) &= (-1)^{\frac{n}{2}} B(x^2), & B\left(\frac{1}{\varkappa^2}\right) &= (-1)^{\frac{n}{2}} \sqrt{\frac{\varkappa'}{\varkappa}}^{n^2-4}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^{n^2} C\left(\frac{z^2}{\varkappa^2 y^2}\right) &= C(x^2), & C\left(\frac{1}{\varkappa^2}\right) &= \sqrt{\frac{\varkappa'}{\varkappa}}^{n^2-4}, \\
\left(y\sqrt{\frac{\varkappa}{\varkappa'}}\right)^2 D\left(\frac{z^2}{\varkappa^2 y^2}\right) &= (-1)^{\frac{n}{2}} D(x^2), & D\left(\frac{1}{\varkappa^2}\right) &= (-1)^{\frac{n}{2}} \sqrt{\frac{\varkappa'}{\varkappa}}^{n^2-4},
\end{aligned} \tag{8}$$

for n even, and

$$\begin{aligned}
\left(y\sqrt{\frac{\kappa}{\kappa'}}\right)^{n^2-1} A\left(\frac{z^2}{\kappa^2 y^2}\right) &= (-1)^{\frac{n-1}{2}} C(x^2), & A\left(\frac{1}{\kappa^2}\right) &= (-1)^{\frac{n-1}{2}} \sqrt{\frac{\kappa'}{\kappa}}^{n^2-1}, \\
\left(y\sqrt{\frac{\kappa}{\kappa'}}\right)^{n^2-1} B\left(\frac{z^2}{\kappa^2 y^2}\right) &= D(x^2), & B\left(\frac{1}{\kappa^2}\right) &= \sqrt{\frac{\kappa'}{\kappa}}^{n^2-1}, \\
\left(y\sqrt{\frac{\kappa}{\kappa'}}\right)^{n^2-1} C\left(\frac{z^2}{\kappa^2 y^2}\right) &= (-1)^{\frac{n-1}{2}} A(x^2), & C\left(\frac{1}{\kappa^2}\right) &= (-1)^{\frac{n-1}{2}} n \sqrt{\frac{\kappa'}{\kappa}}^{n^2-1}, \\
\left(y\sqrt{\frac{\kappa}{\kappa'}}\right)^{n^2-1} D\left(\frac{z^2}{\kappa^2 y^2}\right) &= B(x^2), & D\left(\frac{1}{\kappa^2}\right) &= \sqrt{\frac{\kappa'}{\kappa}}^{n^2-1},
\end{aligned} \tag{9}$$

for n odd.

According to these formulae, in the case of odd n one can reduce the four polynomials $A(x^2), B(x^2), C(x^2), D(x^2)$ to one of them, for example to $A(x^2)$.

If in the formulae I, II one divides the first three formulae by the last, then one obtains the multiplication formulae for the elliptic functions (§ 42):

$$\begin{aligned}
&\text{sn } nv = \frac{xyz A(x^2)}{D(x^2)}, \\
\text{III. For even } n : &\quad \text{cn } nv = \frac{B(x^2)}{D(x^2)}, \\
&\quad \text{dn } nv = \frac{C(x^2)}{D(x^2)}, \\
&\quad \text{sn } nv = \frac{x A(x^2)}{D(x^2)}, \\
\text{IV. For odd } n : &\quad \text{cn } nv = \frac{y B(x^2)}{D(x^2)}, \\
&\quad \text{dn } nv = \frac{z C(x^2)}{D(x^2)}.
\end{aligned}$$

The coefficients of A, B, C, D still depend on ω or on κ^2 . The manner of this dependence is revealed by the recursion formulae which follow from the addition theorem for the successive calculation of A, B, C, D . If the value of the multiplier n to which these functions belong is indicated by an index, then first of all we have

$$A_1 = 1, \quad B_1 = 1, \quad C_1 = 1, \quad D_1 = 1; \tag{10}$$

further, from the addition formulae [§ 44, (16), for $u = v$],

$$\begin{aligned}
\text{sn } 2v &= \frac{2 \text{sn } v \text{ cn } v \text{ dn } v}{1 - \kappa^2 \text{sn}^4 v}, \\
\text{cn } 2v &= \frac{\text{cn}^2 v - \text{sn}^2 v \text{ dn}^2 v}{1 - \kappa^2 \text{sn}^4 v}, \\
\text{dn } 2v &= \frac{\text{dn}^2 v - \kappa^2 \text{sn}^2 v \text{ cn}^2 v}{1 - \kappa^2 \text{sn}^4 v},
\end{aligned} \tag{11}$$

and hence

$$A_2 = 2, \quad B_2 = 1 - 2x^2 + \kappa^2 x^4, \quad C_2 = 1 - 2\kappa^2 x^2 + \kappa^2 x^4, \quad D_2 = 1 - \kappa^2 x^4. \tag{12}$$

If in (11) we replace v by nv , then

$$\begin{aligned}
A_{2n} &= 2A_n B_n C_n D_n, \\
B_{2n} &= B_n^2 D_n^2 - x^2 y^2 z^2 A_n^2 C_n^2, \quad n \text{ even}, \\
&= y^2 B_n^2 D_n^2 - x^2 z^2 A_n^2 C_n^2, \quad n \text{ odd}, \\
C_{2n} &= C_n^2 D_n^2 - \kappa^2 x^2 y^2 z^2 A_n^2 B_n^2, \quad n \text{ even}, \\
&= z^2 C_n^2 D_n^2 - \kappa^2 x^2 y^2 A_n^2 B_n^2, \quad n \text{ odd}, \\
D_{2n} &= D_n^4 - \kappa^2 x^4 y^4 z^4 A_n^4, \quad n \text{ even}, \\
&= D_n^4 - \kappa^2 x^4 A_n^4, \quad n \text{ odd}.
\end{aligned} \tag{13}$$

If in the addition formulae of § 44 one puts $u = nv$, $v = (n+1)v$, then

$$\begin{aligned}
A_{2n+1} &= y^2 z^2 A_n D_n B_{n+1} C_{n+1} + A_{n+1} D_{n+1} C_n B_n, \quad n \text{ even}, \\
&= A_n D_n B_{n+1} C_{n+1} + y^2 z^2 A_{n+1} D_{n+1} C_n B_n, \quad n \text{ odd}, \\
B_{2n+1} &= B_n B_{n+1} D_n D_{n+1} - x^2 z^2 A_n A_{n+1} C_n C_{n+1}, \\
C_{2n+1} &= C_n C_{n+1} D_n D_{n+1} - \kappa^2 x^2 y^2 A_n A_{n+1} B_n B_{n+1}, \\
D_{2n+1} &= D_n^2 D_{n+1}^2 - \kappa^2 x^4 y^2 z^2 A_n^2 A_{n+1}^2.
\end{aligned} \tag{14}$$

From these formulae one concludes that the A, B, C, D are integral rational functions of x^2 , and that the numerical coefficients are integral rational numbers. For this property holds by (10), (12) for $n = 1, 2$, and hence in general by (13), (14).

Concerning the degree with respect to x^2 one may also conclude that the coefficients of $x^{2\nu}$ do not exceed degree ν with respect to κ^2 . For if this rule is correct for n and $n+1$, then its correctness follows for $2n$ and $2n+1$; and it is true for $n = 1, n = 2$.

From the fundamental equations between the three elliptic functions,

$$1 = \text{cn}^2 v + \text{sn}^2 v = \text{dn}^2 v + \kappa^2 \text{sn}^2 v,$$

there also result the relations

$$\begin{aligned}
D^2 &= B^2 + x^2 y^2 z^2 A^2 = C^2 + \kappa^2 x^2 y^2 z^2 A^2, \quad n \text{ even}, \\
D^2 &= y^2 B^2 + x^2 A^2 = z^2 C^2 + \kappa^2 x^2 A^2, \quad n \text{ odd},
\end{aligned} \tag{15}$$

with whose help, from (14), one can express B_{2n} and C_{2n} in terms of A_n and D_n alone:

$$\begin{aligned}
B_{2n} &= D_n^4 - 2x^2 y^2 z^2 A_n^2 D_n^2 + \kappa^2 x^4 y^4 z^4 A_n^4, \\
C_{2n} &= D_n^4 - 2\kappa^2 x^2 y^2 z^2 A_n^2 D_n^2 + \kappa^2 x^4 y^4 z^4 A_n^4, \quad n \text{ even},
\end{aligned} \tag{16a}$$

$$\begin{aligned}
B_{2n} &= D_n^4 - 2x^2 A_n^2 D_n^2 + \kappa^2 x^4 A_n^4, \\
C_{2n} &= D_n^4 - 2\kappa^2 x^2 A_n^2 D_n^2 + \kappa^2 x^4 A_n^4, \quad n \text{ odd}.
\end{aligned} \tag{16b}$$

§ 58. Multiplication of the function $\wp(u)$.

A very elegant form is taken by the multiplication theory for the Weierstrass function $\wp(u)$ with the two periods ω_1, ω_2 .

Consider the doubly periodic function

$$\wp(nu) - \wp(u), \tag{1}$$

where n may be any positive integer. This function becomes infinite for $u = 0$, and in such a way that

$$\wp(nu) - \wp(u) + \frac{n^2 - 1}{n^2} \frac{1}{u^2} \tag{2}$$

remains finite for $u = 0$. But it also becomes infinite for all values u^* of u satisfying

$$nu^* \equiv 0 \pmod{\omega_1, \omega_2},$$

that is, values of the form

$$u^* = \frac{\nu_1 \omega_1 + \nu_2 \omega_2}{n}, \quad (3)$$

where ν_1, ν_2 denote arbitrary integers. We obtain all incongruent values contained in (3) by letting ν_1 and ν_2 each run through a complete system of residues modulo n .

The function (1) vanishes for all non-zero values u^0 of u which satisfy

$$\pm nu^0 \equiv u^0 \pmod{\omega_1, \omega_2},$$

that is, for

$$u^0 = \frac{\nu_1 \omega_1 + \nu_2 \omega_2}{n \pm 1}, \quad (4)$$

again with arbitrary integers ν_1, ν_2 . The zeros are of first order and the poles of second order.

We therefore introduce a function $\psi_n(u)$, defined as follows:

$$\psi_n(u)^2 = n^2 \prod^{\nu_1, \nu_2} \left[\wp(u) - \wp\left(\frac{\nu_1 \omega_1 + \nu_2 \omega_2}{n}\right) \right], \quad (5)$$

where ν_1, ν_2 each pass through a complete residue system modulo n , with only the pair $0, 0$ excluded, and we add the further determination that $\nu_1 = 1$ is to hold.

When n is odd, each linear factor occurs twice in the product (5), since the two incongruent values

$$u = \pm \frac{\nu_1 \omega_1 + \nu_2 \omega_2}{n}$$

give the same value of $\wp(u)$. But when n is even, all linear factors again occur twice in (5), with the exception of the three

$$\wp(u) - \wp\left(\frac{\omega_1}{2}\right), \quad \wp(u) - \wp\left(\frac{\omega_2}{2}\right), \quad \wp(u) - \wp\left(\frac{\omega_1 + \omega_2}{2}\right),$$

which occur only simply. If one observes now that, by § 41,

$$\wp'(u)^2 = 4 \left[\wp(u) - \wp\left(\frac{\omega_1}{2}\right) \right] \left[\wp(u) - \wp\left(\frac{\omega_2}{2}\right) \right] \left[\wp(u) - \wp\left(\frac{\omega_1 + \omega_2}{2}\right) \right],$$

then the result is

$$n \text{ odd : } \psi_n = P_n, \quad n \text{ even : } \psi_n = \wp'(u) P_n, \quad (6)$$

where P_n denotes an integral rational function of $\wp(u)$, of degree $\frac{1}{2}(n^2 - 1)$ for odd n and of degree $\frac{1}{2}(n^2 - 4)$ for even n , and whose highest term is respectively

$$n \wp(u)^{\frac{n^2-1}{2}}, \quad -\frac{n}{2} \wp(u)^{\frac{n^2-4}{2}}.$$

This also determines the sign which was still undetermined in (5). With this determination of the signs the initial term in the expansion of $\psi_n(u)$ in ascending powers of u is in both cases (§ 56)

$$\frac{n}{u^{n^2-1}}. \quad (7)$$

Now observe that the poles of the function (1) coincide with the zeros of $\psi_n(u)$, and the zeros of (1) with the zeros of $\psi_{n+1}(u)$; consequently

$$\frac{\psi_n(u)^2 [\wp(nu) - \wp(u)]}{\psi_{n+1}(u) \psi_{n-1}(u)} \quad (8)$$

is an everywhere finite doubly periodic function, and hence a constant, whose value at $u = 0$ is found to be -1 . Thus

$$\wp(nu) = \wp(u) - \frac{\psi_{n+1}(u)\psi_{n-1}(u)}{\psi_n(u)^2}, \quad (9)$$

from which the two following formulae follow:

$$\begin{aligned} \wp(nu) &= \wp(u) - \frac{P_{n+1}P_{n-1}}{\wp'(u)^2 P_n^2}, \quad n \text{ even}, \\ &= \wp(u) - \frac{\wp'(u)^2 P_{n+1}P_{n-1}}{P_n^2}, \quad n \text{ odd}. \end{aligned} \quad (10)$$

We first determine the function P_n in the first cases $n = 1, 2, 3, 4$. For this purpose, by the addition theorem of the $\wp$ -function [§ 49, (11)], we form

$$4[\wp(u) + \wp(v) + \wp(u+v)] = \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2,$$

and put $u = v$, determining the limit on the right by differentiation:

$$\wp(2u) + 2\wp(u) = \frac{1}{4} \left(\frac{\wp''(u)}{\wp'(u)} \right)^2 = \frac{\frac{1}{4}(6\wp(u)^2 - \frac{1}{2}g_2)^2}{4\wp(u)^3 - g_2\wp(u) - g_3},$$

or

$$\wp(2u) = \wp(u) - \frac{3\wp(u)^4 - \frac{3}{2}g_2\wp(u)^2 - 3g_3\wp(u) - \frac{g_2^2}{16}}{\wp'(u)^2}. \quad (11)$$

Therefore

$$\psi_1 = 1, \quad \psi_2 = -\wp'(u), \quad \psi_3 = P_3,$$

and

$$P_1 = 1, \quad P_2 = -1, \quad P_3 = 3\wp(u)^4 - \frac{3}{2}g_2\wp(u)^2 - 3g_3\wp(u) - \frac{g_2^2}{16}. \quad (12)$$

Furthermore, from the addition formula (§ 49),

$$\wp(v+u) - \wp(v-u) = -\frac{\wp'(u)\wp'(v)}{[\wp(v) - \wp(u)]^2},$$

by putting $v = 2u$ we obtain, using (9),

$$\wp(3u) - \wp(u) = -\frac{\wp'(u)\wp'(2u)}{[\wp(2u) - \wp(u)]^2} = -\frac{\wp'(u)^5\wp'(2u)}{\psi_3(u)^2} = -\frac{\psi_2(u)\psi_4(u)}{\psi_3(u)^2},$$

hence

$$\psi_4(u) = \wp'(u)P_4 = -\wp'(u)^4\wp'(2u).$$

If now the value $\wp'(2u)$ is formed from (11), one obtains

$$\begin{aligned} P_4 &= -2\wp(u)^6 + \frac{5}{2}g_2\wp(u)^4 + 10g_3\wp(u)^3 + \frac{5}{8}g_2^2\wp(u)^2 \\ &\quad + \frac{1}{2}g_2g_3\wp(u) + g_3^2 - \frac{g_2^3}{32}. \end{aligned} \quad (13)$$

For larger values of n we derive recursion formulae. To this end we apply (9) to two different numbers m, n and obtain

$$\wp(u) - \wp(mu) = \frac{\psi_{m+1}(u)\psi_{m-1}(u)}{\psi_m(u)^2}, \quad \wp(u) - \wp(nu) = \frac{\psi_{n+1}(u)\psi_{n-1}(u)}{\psi_n(u)^2}.$$

From this one sees that the right hand sides become equal for those values u^0 of u which are different from zero and satisfy

$$mu^0 \equiv \pm nu^0 \pmod{\omega_1, \omega_2},$$

or

$$(m \pm n)u^0 \equiv 0 \pmod{\omega_1, \omega_2}.$$

For the same values u^0 the functions

$$\psi_{m \pm n}(u)$$

also vanish, so that the two functions

$$\psi_{m+1}(u)\psi_{m-1}(u)\psi_n(u)^2 - \psi_{n+1}(u)\psi_{n-1}(u)\psi_m(u)^2, \quad \psi_{m+n}(u)\psi_{m-n}(u),$$

which by (6) are integral rational functions of $\wp(u)$, vanish for the same finite values of $\wp(u)$. They therefore differ only by a constant factor, which is obtained from (7) as equal to 1. We have, therefore,

$$\psi_{m+n}(u)\psi_{m-n}(u) = \psi_{m+1}(u)\psi_{m-1}(u)\psi_n(u)^2 - \psi_{n+1}(u)\psi_{n-1}(u)\psi_m(u)^2. \quad (14)$$

We apply this formula to two special cases by putting $n+1, n$ or $n+1, n-1$ in place of m, n , and obtain

$$\begin{aligned} \psi_{2n+1}(u) &= \psi_{n+2}(u)\psi_n(u)^3 - \psi_{n+1}(u)^3\psi_{n-1}(u), \\ \wp'(u)\psi_{2n}(u) &= -\psi_n(u)[\psi_{n+2}(u)\psi_{n-1}(u)^2 - \psi_{n+1}(u)^2\psi_{n-2}(u)]. \end{aligned}$$

Hence, by (6), the recursion formulae for P_n are

$$\begin{aligned} P_{2n+1} &= \wp'(u)^4 P_{n+2} P_n^3 - P_{n+1}^3 P_{n-1}, \quad n \text{ even}, \\ &= P_{n+2} P_n^3 - \wp'(u)^4 P_{n+1}^3 P_{n-1}, \quad n \text{ odd}, \end{aligned} \quad (15)$$

$$P_{2n} = -P_n(P_{n+2}P_{n-1}^2 - P_{n+1}^2P_{n-2}). \quad (16)$$

Since all the functions P_n can thus be calculated from P_1, P_2, P_3, P_4 , it follows that the coefficients of P_n are composed rationally from g_2, g_3 and rational numbers.

Our further considerations on division, however, will be attached to the multiplication formulae of the functions $\operatorname{sn} u, \operatorname{cn} u, \operatorname{dn} u$, whose advantages will come into full force only in the fourth part.

§ 59. Division by 2.

Turning first to the simplest case, we put v in place of $2v$ in the equations (5) of § 57. If

$$x = \operatorname{sn} \frac{v}{2}, \quad y = \operatorname{cn} \frac{v}{2}, \quad z = \operatorname{dn} \frac{v}{2}, \quad (1)$$

then

$$\begin{aligned} \operatorname{sn} v &= \frac{2xyz}{1 - \kappa^2 x^4}, \\ \operatorname{cn} v &= \frac{y^2 - x^2 z^2}{1 - \kappa^2 x^4}, \\ \operatorname{dn} v &= \frac{z^2 - \kappa^2 x^2 y^2}{1 - \kappa^2 x^4}. \end{aligned} \quad (2)$$

The last two of these equations are quadratic with respect to x^2 , but they have only one common root; for the roots of the second equation (2) are

$$\operatorname{sn}^2 \frac{v}{2}, \quad \operatorname{sn}^2 \left(\frac{v}{2} + K + iK' \right),$$

and those of the third are

$$\operatorname{sn}^2 \frac{v}{2}, \quad \operatorname{sn}^2 \left(\frac{v}{2} + K \right).$$

From (2) one now easily finds

$$\begin{aligned} 1 + \operatorname{cn} v &= \frac{2y^2}{1 - \kappa^2 x^4}, & 1 + \operatorname{dn} v &= \frac{2z^2}{1 - \kappa^2 x^4}, \\ 1 - \operatorname{cn} v &= \frac{2x^2 z^2}{1 - \kappa^2 x^4}, & 1 - \operatorname{dn} v &= \frac{2\kappa^2 x^2 y^2}{1 - \kappa^2 x^4}, \\ \operatorname{dn} v + \operatorname{cn} v &= \frac{2y^2 z^2}{1 - \kappa^2 x^4}. \end{aligned}$$

Thus

$$\begin{aligned} x &= \sqrt{\frac{1 - \operatorname{cn} v}{1 + \operatorname{dn} v}} = \frac{1}{\kappa} \sqrt{\frac{1 - \operatorname{dn} v}{1 + \operatorname{cn} v}}, \\ y &= \sqrt{\frac{\operatorname{dn} v + \operatorname{cn} v}{1 + \operatorname{dn} v}}, \quad z = \sqrt{\frac{\operatorname{dn} v + \operatorname{cn} v}{1 + \operatorname{cn} v}}. \end{aligned} \tag{3}$$

Between the signs of these three quantities there is still, by the first equation (2), a relation, so that one obtains only four different systems of values, having the following meaning:

$$\begin{array}{lll} \operatorname{sn}(\frac{v}{2}), & \operatorname{cn}(\frac{v}{2}), & \operatorname{dn}(\frac{v}{2}), \\ \operatorname{sn}(\frac{v}{2} + 2K), & \operatorname{cn}(\frac{v}{2} + 2K), & \operatorname{dn}(\frac{v}{2} + 2K), \\ \operatorname{sn}(\frac{v}{2} + 2iK'), & \operatorname{cn}(\frac{v}{2} + 2iK'), & \operatorname{dn}(\frac{v}{2} + 2iK'), \\ \operatorname{sn}(\frac{v}{2} + 2K + 2iK'), & \operatorname{cn}(\frac{v}{2} + 2K + 2iK'), & \operatorname{dn}(\frac{v}{2} + 2K + 2iK'). \end{array}$$

We also give the special cases following from (3):

$$\begin{aligned} \operatorname{sn} \frac{K}{2} &= \frac{1}{\sqrt{1 + \kappa'}}, & \operatorname{sn} \frac{iK'}{2} &= \frac{i}{\sqrt{\kappa}}, & \operatorname{sn} \frac{K + iK'}{2} &= \sqrt{\frac{\kappa + i\kappa'}{\kappa}}, \\ \operatorname{cn} \frac{K}{2} &= \sqrt{\frac{\kappa'}{1 + \kappa'}}, & \operatorname{cn} \frac{iK'}{2} &= \sqrt{\frac{1 + \kappa}{\kappa}}, & \operatorname{cn} \frac{K + iK'}{2} &= \sqrt{\frac{-i\kappa'}{\kappa}}, \\ \operatorname{dn} \frac{K}{2} &= \sqrt{\kappa'}, & \operatorname{dn} \frac{iK'}{2} &= \sqrt{1 + \kappa}, & \operatorname{dn} \frac{K + iK'}{2} &= \sqrt{\frac{\kappa}{\kappa + i\kappa'}}. \end{aligned}$$

It follows now that division by 2, and consequently also by every power of 2, can be carried out through a chain of square roots. If therefore the problem of division by an odd number is assumed solved, then division by an even number has been reduced to square roots. In what follows we shall concern ourselves exclusively with division by odd numbers.

§ 60. Division by an odd number.

If n is an odd number and

$$x = \operatorname{sn} \frac{v}{n}, \quad y = \operatorname{cn} \frac{v}{n}, \quad z = \operatorname{dn} \frac{v}{n}, \tag{1}$$

then from equations IV, § 57, we get

$$D(x^2) \operatorname{sn} v - xA(x^2) = 0, \quad D(x^2) \operatorname{cn} v - yB(x^2) = 0, \quad D(x^2) \operatorname{dn} v - zC(x^2) = 0. \tag{2}$$

The first of these equations is of degree n^2 with respect to the unknown x ; by the second and third, y and z are expressed rationally in x (and in $\text{cn } v, \text{dn } v$). Thus there arise n^2 systems of values for the three unknowns x, y, z , with the meaning

$$\begin{aligned} x_{\mu, \mu'} &= \text{sn} \left(\frac{v}{n} + \frac{4\mu K + 4\mu' i K'}{n} \right), \\ y_{\mu, \mu'} &= \text{cn} \left(\frac{v}{n} + \frac{4\mu K + 4\mu' i K'}{n} \right), \\ z_{\mu, \mu'} &= \text{dn} \left(\frac{v}{n} + \frac{4\mu K + 4\mu' i K'}{n} \right), \end{aligned} \quad (3)$$

where μ, μ' each run through a complete system of residues modulo n .

The first equation (2),

$$D(x^2) \text{sn } v - xA(x^2) = 0, \quad (4)$$

of degree n^2 , is called the general division equation. It is irreducible in the sense that it cannot be decomposed into factors which are rational with respect to $\text{sn } v, \text{cn } v, \text{dn } v$ and have arbitrary coefficients independent of v .

For, first, the n^2 values $x_{\mu, \mu'}$ are all different from one another; and if any rational equation

$$F \left(\text{sn } \frac{v}{n}, \text{sn } v, \text{cn } v, \text{dn } v \right) = 0$$

holds, then the variable v in it can be replaced by $v + 4\mu K + 4\mu' i K'$. Because of the periodicity of $\text{sn } v, \text{cn } v, \text{dn } v$, it follows that

$$F(x, \text{sn } v, \text{cn } v, \text{dn } v) = 0$$

is satisfied for all roots of (4). To determine its Galois group, we first fix the field of rationality. It is to comprise the following quantities:

1. rational numbers,
2. rational functions of $\varkappa^2$,
3. the three functions $\text{sn } v, \text{cn } v, \text{dn } v$,
4. the quantities $\text{sn} \left(\frac{4\mu K}{n} + \frac{4\mu' i K'}{n} \right)$.

Then from (2) it follows that

$$\text{cn} \left(\frac{4\mu K + 4\mu' i K'}{n} \right), \quad \text{dn} \left(\frac{4\mu K + 4\mu' i K'}{n} \right)$$

also belong to the field of rationality.

We note now that, by the addition theorem, each of the quantities $x_{\mu, \mu'}$ can be expressed rationally by any other among them. If in particular

$$x_{\mu, \mu'} = F_{\mu, \mu'}(x_{0,0}), \quad (5)$$

then

$$x_{\mu+\nu, \mu'+\nu'} = F_{\mu, \mu'}(x_{\nu, \nu'}) = F_{\mu+\nu, \mu'+\nu'}(x_{0,0}). \quad (6)$$

It follows that the Galois group of our equation consists of the n^2 substitutions $S_{\nu, \nu'}$ obtained by increasing the indices μ, μ' in the roots $x_{\mu, \mu'}$ all by the same pair of numbers ν, ν' (where each index is to be taken modulo n).

Indeed, by (5) every rational function of the roots $x_{\mu, \mu'}$ can be expressed rationally by $x_{0,0}$ in the form

$$\Phi(x_{0,0})$$

and hence, by (6), goes under the substitution $S_{\nu,\nu'}$ into $\Phi(x_{\nu,\nu'})$. If it remains unaltered by this substitution, it is therefore rational. Conversely, it follows from irreducibility that every rational equation between the roots, since it can be put into the form

$$\psi(x_{0,0}) = 0,$$

and consequently implies

$$\psi(x_{\nu,\nu'}) = 0,$$

admits the substitution $S_{\nu,\nu'}$. But the group of the $S_{\nu,\nu'}$ is, because of

$$S_{\nu,\nu'}S_{\mu,\mu'} = S_{\nu+\mu,\nu'+\mu'},$$

an Abelian group, which, according to the formula

$$S_{\nu,\nu'} = S_{1,0}^\nu S_{0,1}^{\nu'},$$

is represented by the basis $S_{1,0}, S_{0,1}$; hence the general division equation is an Abelian equation and therefore algebraically solvable (Vol. I, § 169 f.).

§ 61. Division of the periods.

The problem still to be solved consists now in determining algebraically the quantities

$$x_{\mu,\mu'} = \operatorname{sn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right). \quad (1)$$

One obtains all values of this quantity by letting μ, μ' , independently of one another, each run through a complete system of residues modulo n . These values, however, are also all different from one another; for $\operatorname{sn} v$ can be equal to $\operatorname{sn} v'$ only when v' is congruent to v , or congruent to $2K - v$, modulo $4K, 2iK'$, and both cannot occur for two different arguments of (1).

The n^2 quantities (1) are the roots of the equation

$$xA(x^2) = 0, \quad (2)$$

and from the formulae

$$y = \frac{D(x^2)}{B(x^2)}, \quad z = \frac{D(x^2)}{C(x^2)} \quad (3)$$

the quantities

$$y_{\mu,\mu'} = \operatorname{cn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right), \quad z_{\mu,\mu'} = \operatorname{dn}\left(\frac{4\mu K + 4\mu' iK'}{n}\right)$$

can also be rationally expressed by $x_{\mu,\mu'}$, if the field of rationality consists of rational numbers and rational functions of $\varkappa^2$.

From the roots x^2 of the equation $A = 0$, the roots of $B = 0$, $C = 0$, $D = 0$ can be rationally derived according to § 60, (2), and § 44, (18), (19), (20). Namely, if x^2 is a root of $A = 0$, then

$$\frac{1 - x^2}{1 - \varkappa^2 x^2}, \quad \frac{1 - \varkappa^2 x^2}{\varkappa^2(1 - x^2)}, \quad \frac{1}{\varkappa^2 x^2}$$

are the roots of $B = 0$, $C = 0$, $D = 0$. The meaning of these latter roots is

$$\operatorname{sn}\left(\frac{(4\mu + 1)K + 4\mu' iK'}{n}\right)^2, \\ \operatorname{sn}\left(\frac{(4\mu + 1)K + (4\mu' + 1)iK'}{n}\right)^2,$$

$$\operatorname{sn}\left(\frac{4\mu K + (4\mu' + 1)iK'}{n}\right)^2.$$

Thus, by solving the equation $A = 0$, all quantities of the form

$$\operatorname{sn}\left(\frac{\mu K + \mu' iK'}{n}\right)^2,$$

where μ, μ' are arbitrary integers, are rationally determined.

§ 62. Abelian relations.

For a closer investigation of the algebraic nature of the period-division equation, a system of relations between its roots is of great importance; we now turn to its derivation.¹

We consider the sum

$$\sum_{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(u + \frac{2\nu}{n}\right)}{\vartheta_{g_1, g_2}\left(u + \frac{2\nu}{n}\right)}, \quad (1)$$

where ν is taken over a complete system of residues for the (odd) modulus n . Here ν' is an arbitrary integer and g_1, g_2 is one of the three even characteristics $(0, 0), (0, 1), (1, 0)$. This sum is independent of the particular residue system chosen for ν .

The common denominator of the fractions occurring in (1),

$$\prod_{\nu} \vartheta_{g_1, g_2}\left(u + \frac{2\nu}{n}\right) = \pm \prod_{\nu} \vartheta_{g_1, g_2}\left(u + \frac{\nu}{n}\right),$$

is, apart from a constant factor, equal by the last formulae of § 33 to

$$\vartheta_{g_1, g_2}(nu, n\omega).$$

Thus, if we put the sum (1) equal to

$$\frac{\Phi(u)}{\vartheta_{g_1, g_2}(nu, n\omega)}, \quad (2)$$

then $\Phi(u)$ is an integral transcendental function of u . Now

$$\frac{\vartheta_{11}\left(u + \frac{2\nu+1}{n}\right)}{\vartheta_{g_1, g_2}\left(u + \frac{2\nu+1}{n}\right)} = (-1)^{g_1+1} \frac{\vartheta_{11}\left(u + \frac{2\nu+n+1}{n}\right)}{\vartheta_{g_1, g_2}\left(u + \frac{2\nu+n+1}{n}\right)}$$

and, simultaneously with ν , the integer $\nu + \frac{1}{2}(n+1)$ runs through a complete system of residues modulo n . Hence from (1) and (2) there follows

$$\Phi\left(u + \frac{1}{n}\right) = -e^{-\frac{4\nu'\pi i}{n}} \Phi(u), \quad (3)$$

and similarly

$$\Phi(u + \omega) = -e^{-n\pi i(2u+\omega)} \Phi(u). \quad (4)$$

It follows that $\Phi(u)$ is a t -function with periods $1/n, \omega$, determined up to a factor independent of u by the conditions (3), (4), and expressible by a ϑ -function (§ 19). It is easy to see that the conditions (3), (4) are satisfied by the function

$$e^{-4\pi i\nu'u} \vartheta_{11}(nu - 4\nu'\omega, n\omega),$$

¹These relations originate with Abel (Oeuvres complètes, vol. I, p. 523; vol. II, p. 251 of the new edition). The proof of these relations given above follows Hermite (Crelle's Journal, vol. 32, p. 283). Also to be mentioned are Sylow, Christiania Videnskabselskabs Forhandling 1864 and 1871; Kronecker, Reports of the Berlin Academy, 19 July 1875; Engel, Reports of the Saxon Society of Sciences, 31 July 1884.

and hence

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(u + \frac{2\nu}{n}\right)}{\vartheta_{g_1, g_2}\left(u + \frac{2\nu}{n}\right)} = C e^{-4\pi i \nu' u} \frac{\vartheta_{11}(nu - 2\nu'\omega, n\omega)}{\vartheta_{g_1, g_2}(nu, n\omega)}. \quad (5)$$

The constant C can be determined easily if one multiplies on the right and on the left by $\vartheta_{g_1, g_2}(u)$ and then puts

$$u = \frac{(1 - g_2) + (1 - g_1)\omega}{2}.$$

Formula (2) of § 21 then gives

$$C = (-1)^{(g_1+1)(g_2+1)\frac{n-1}{2}} n \frac{\vartheta_{g_1, g_2}(0, \omega)}{\vartheta_{g_1, g_2}(2\nu'\omega, n\omega)} \frac{\vartheta'_{11}(0, n\omega)}{\vartheta'_{11}(0, \omega)}. \quad (6)$$

The Abelian relations are obtained simply by putting $nu = 2\nu'\omega$ in (5), whereby the right hand side vanishes:

$$\sum^{\nu} e^{\frac{2\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(\frac{2\nu'\omega + 2\nu}{n}\right)}{\vartheta_{g_1, g_2}\left(\frac{2\nu'\omega + 2\nu}{n}\right)} = 0. \quad (7)$$

If a linear transformation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad \alpha\delta - \beta\gamma = 1$$

is applied to the ϑ -functions occurring here, then by § 39 one obtains the general formula

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} \frac{\vartheta_{11}\left(\frac{2(\nu\alpha + \nu'\gamma) + 2(\nu\beta + \nu'\delta)\omega}{n}\right)}{\vartheta_{g_1, g_2}\left(\frac{2(\nu\alpha + \nu'\gamma) + 2(\nu\beta + \nu'\delta)\omega}{n}\right)} = 0. \quad (8)$$

If now $g_1, g_2 = 0, 1$ is put and the notation of § 42 is introduced, these equations pass immediately into relations between the roots $x_{\nu, \nu'}$ (§ 61):

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu\alpha + \nu'\gamma, \nu\beta + \nu'\delta} = 0, \quad (9)$$

and, corresponding to the two special cases

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

one obtains

$$\sum^{\nu} e^{\frac{8\nu\nu'\pi i}{n}} x_{\nu, \nu'} = 0, \quad (10)$$

$$\sum^{\nu'} e^{-\frac{8\nu'\nu\pi i}{n}} x_{\nu, \nu'} = 0. \quad (11)$$

From (5), (6) one can also determine the value of these sums when the n th root of unity

$$e^{\frac{8\pi i}{n}}$$

is replaced by another; this is done by replacing ν' in (5) by another symbol, m , and then putting $nu = 2\nu'\omega$. We restrict ourselves to the case $(g_1, g_2) = (0, 1)$ and multiply (5) also by $\vartheta_{00} : \vartheta_{10}$; then

$$\sum^{\nu} e^{\frac{8m\nu\pi i}{n}} x_{\nu, \nu'} = (-1)^{\frac{n-1}{2}} n e^{-\frac{4\pi i m \nu' \omega}{n}} \frac{\vartheta_{00} \vartheta_{01} \vartheta'_{11}(0, n\omega) \vartheta_{11}(2(\nu' - m)\omega, n\omega)}{\vartheta_{10} \vartheta'_{11} \vartheta_{01}(2m\omega, n\omega) \vartheta_{01}(2\nu'\omega, n\omega)}, \quad (12)$$

where the left hand side vanishes then, but also only then, when $m \equiv \nu' \pmod{n}$.